

# Mohammed Eid Abdelmeguid

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Portfolio | GitHub | LinkedIn | Hugging Face

## Summary

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Early-career AI/ML Engineer who builds end-to-end AI systems — retrieval-augmented platforms, Arabic NLP pipelines, and multimodal models — on a software engineering foundation in Python, Django, FastAPI and PostgreSQL. Comfortable across the ML lifecycle, from data ingestion and feature engineering through evaluation and deployment. Currently completing a 9-month AI & Data Science diploma on an MCIT scholarship.

## Selected Projects

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**AI Compass** — Multi-Agent AI News & Recommendation Platform May 2026 – Present

- Led a team of five as team lead and main developer on an AI news platform that pulls content from 7 sources, enriches it through a 5-agent pipeline, and recommends what each reader should see next.
- Built the recommendation engine — pgvector similarity search, weighted scoring and MMR diversification — and measured it with an offline NDCG@10 / MAP harness rather than guessing at quality.
- Designed the semantic search and RAG chat on local Sentence-Transformer embeddings, PostgreSQL/pgvector and Groq-served LLMs, with Celery and Redis running background work across three queues.
- Traced duplicate task runs to an 88-minute pipeline job outliving Redis's one-hour visibility timeout; raised the timeout and moved long batch work onto its own queue so search and chat stayed responsive.

**Palestinian Cultural Knowledge Platform** — Arabic NLP, RAG & Knowledge Graph Jun – Aug 2026

- Built a platform with one teammate that turns a quality-scored, deduplicated Arabic/English corpus on Palestinian culture into searchable knowledge; I owned the ingestion pipeline, Arabic NLP, retrieval and the knowledge graph.
- Arabic named-entity recognition with CAMEL Tools reached F1 0.47, and the RAG retriever reached Recall@5 0.93 across 1,282 indexed passages.
- Published the corpus — 882 documents, roughly 890K words — as an open bilingual dataset on Hugging Face, and built a knowledge graph of 13,063 entities linked against Wikidata.

**AynVQA-MSA** — Arabic Spoken Visual Question Answering Jul – Aug 2026

- Built a pipeline for the ImageEval 2026 / ArabicNLP shared task that hears a spoken Arabic question about an image, recovers its three answer options, and picks the right one: Whisper transcription, schema-constrained parsing, a repair step for bad transcripts, then a vision-language model.
- Scored **79.00% accuracy (395/500)** on the official development split, against the organizers' 39.8% baseline, after the repair step lifted an earlier 74.80% result.
- Replaced the baseline's regex answer parsing with JSON-schema-constrained decoding, and tested chain-of-thought and few-shot prompting before deciding the gains didn't justify shipping either.

**Fitness Activity Tracker** — ML-Powered Exercise Classification Mar – Apr 2026

- Classified barbell exercises and counted repetitions from wrist-worn accelerometer and gyroscope data; handled preprocessing, feature engineering (Chauvenet outlier removal, PCA, FFT features) and evaluation on a 3-person team.
- Compared 5 classifiers across 5 feature sets with grid search; Random Forest reached roughly 99% accuracy on held-out test data.

## Technical Skills

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**Machine Learning:** Scikit-learn, Feature Engineering, Model Evaluation, Grid Search, XGBoost  
**Deep Learning:** PyTorch, TensorFlow / Keras, Neural Networks, Computer Vision  
**LLMs & RAG:** Retrieval-Augmented Generation, Embeddings, pgvector, Ollama, Groq API, Hugging Face, Vision-Language Models (Qwen-VL), Structured Outputs (JSON Schema)  
**NLP & Speech:** Arabic NLP (CAMEL Tools), Named Entity Recognition, Topic Modeling (BERTopic), Knowledge Graphs (NetworkX, Wikidata), Speech-to-Text (Whisper)  
**Backend & Data:** Python, SQL, PostgreSQL, Django, FastAPI, Celery, Redis, SQLAlchemy, REST APIs, Pandas, NumPy  
**Tools:** Docker, Git, GitHub Actions, Linux, Jupyter, Streamlit

## Experience

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- Freelance Backend Developer** — StockGo (Qatar) Nov 2024 – Apr 2025
- Built and deployed a Django REST API and analytics dashboard for a live inventory-tracking product (the-stockgo.com), covering the data model, endpoints and reporting queries over PostgreSQL.
- Backend Developer Intern** — Venture Appital Oct 2024 – Feb 2025
- Built modular REST microservices in Flask with full unit-test coverage.
- Full Stack Developer Intern** — NfertiX Mar 2025 – Apr 2025
- Built an internal Django/JavaScript web application to streamline day-to-day operations.

## Education

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- Faculty of Engineering, Alexandria University** Sep 2019 – Jun 2024
- BS in Communication and Electronics Engineering
- Graduation project: **Advanced Driver Assistance Systems (ADAS)** — Grade A+, ranked 5th of about 60 projects in the Value competition. Team lead on the software side of a five-repository embedded and computer-vision system, mentored by Valeo and sponsored by ITIDA's Egypt Makes Electronics program.

## Certifications

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- AI & Data Science Diploma** — Digilians (MCIT / Egyptian Military Academy scholarship) Dec 2025 – Sep 2026
- IBM AI Engineering Professional Certificate** — IBM / Coursera Jun 2026
- Machine Learning Specialization** — Stanford & DeepLearning.AI (Andrew Ng) May 2026
- Generative AI for Data Science with Copilot** — Microsoft / Coursera Apr 2026
- Microsoft Data Engineer (159 hours)** — Digital Egypt Pioneers Initiative (DEPI) Apr – Oct 2024
- Preparing for **AWS Certified Machine Learning Engineer – Associate (MLA-C01)**